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Picking Up Nickels in Front of a Steamroller: Could LTCM's Risk Model Have Seen 1998 Coming?

A pre-registered backtest of value-at-risk models on synthetic Long-Term Capital Management trades, 1995–2025

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Abstract

Long-Term Capital Management (LTCM) measured risk with volatilities and correlations estimated from recent history and treated returns as approximately normal. This study asks whether a risk model of that kind would have warned of the 1998 losses on LTCM-style trades. Three characteristic trades, short equity volatility, credit convergence and liquidity convergence, are rebuilt from free public data and combined into an equal-risk portfolio targeting 15% annual volatility. Five risk models, from a normal model estimated on five years of data to GARCH-t filtered historical simulation and extreme value theory, forecast the portfolio's one-day value-at-risk on 7,654 trading days from January 1995 to May 2025. The hypotheses, tests, decision rules, a power analysis and 180 robustness specifications were fixed in a pre-registration plan before any results were computed; the few later changes are reported. The LTCM-style model is exceeded on 173 days where 77 were expected (2.26% against 1%), its exceedances cluster, and the primary pre-registered backtest rejects it ($p < 0.001$). Between the Russian default and the week after the rescue, the portfolio lost 25.7% of its capital; the worst day was an 8.9 standard-deviation loss, and the model gave the cumulative loss a probability below one in a million, although its own estimation window contained a larger one-day loss. Better models help but none of the pre-registered ones passes: GARCH-t filtered historical simulation comes closest, at 1.23%, but is narrowly rejected by the pre-registered one-sided test, while a GARCH model with an extreme-value tail, added after the results were seen, passes the 99% VaR backtest, though not the 95% one. All models understate the size of losses beyond the VaR. Correlations between the trades rose from 0.05 to 0.42 in 1998, an increase that survives a post-hoc adjustment for higher volatility, but not in 2008 or 2020, when they were already moderately positive. The portfolio reproduces LTCM's losses in August and September 1998 but not in May and June, so the results describe the market-wide crash in LTCM-style exposures rather than the fund's whole book.

Keywords: Long-Term Capital Management, value-at-risk, backtesting, fat tails, convergence trading, limits to arbitrage, pre-registration

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1. Introduction

On Friday 21 August 1998, four days after Russia defaulted on its domestic debt, Long-Term Capital Management (LTCM) lost $553 million, about 15% of its capital, in a single trading session. By the fund's own calculations, it was unlikely to lose more than $35 million on any single day (Lowenstein, 2000, p. 147), a figure of the kind now called value-at-risk (VaR). By the end of September, a fund founded by some of the most celebrated traders and financial economists of its generation had lost more than 90% of its capital, and the Federal Reserve Bank of New York had brought fourteen banks together to take it over. Four years earlier, LTCM had told investors that a loss of 20% or more in a year should happen about once every fifty years.

The episode is usually told as a story about leverage and hubris. It is also a story about a risk model. LTCM measured risk as volatility estimated from recent history, treated returns as approximately normal, and relied on the diversification of trades that looked unrelated in normal times (Jorion, 2000; Lowenstein, 2000). Each of these assumptions has been criticised since, and later models, from exponentially weighted volatility to extreme value theory, were developed partly in response. What is less often done is to test the LTCM approach in the way that any risk model should be tested: against a long record of out-of-sample forecasts, with a formal backtest, and against the alternatives that a careful risk manager could have used.

This paper asks a narrow question: would a risk model of the kind LTCM used have warned of the 1998 losses on LTCM-style trades, and would better models or the behaviour of the same trades in later crises tell a different story? Because LTCM's positions were never published, the study rebuilds three of its characteristic trade types from free public data: selling equity volatility, holding corporate bonds against Treasuries, and holding older, less liquid Treasuries against the newest issue. A fourth trade, the US swap spread, can only be built from 2000 onwards and is used in the later crises. The three trades are combined into an equal-risk portfolio scaled to 15% annual volatility, the level of the stock market that LTCM said it aimed for. Five risk models then forecast the portfolio's one-day VaR on each of 7,654 trading days from January 1995 to May 2025.

The design follows the argument of Ioannidis (2005) that the credibility of a finding depends on how much freedom the researcher had to shape it. Every hypothesis, test, decision rule and robustness choice was written into a pre-registration plan before any results were computed. The plan names one primary test, reports its statistical power, defines in advance what counts as support, contradiction or an inconclusive outcome, and commits to reporting all 180 robustness specifications whatever they show.

The results are clear on the main question. The LTCM-style model, a normal distribution with volatilities and correlations estimated from the previous five years, is exceeded on 173 days where 77 were expected, an exceedance rate of 2.26% against the promised 1%, and its exceedances cluster in time. The pre-registered primary test rejects it decisively. In the 32 trading days from the Russian default to the week after the rescue, the synthetic portfolio lost

25.7% of its capital. The worst day, 27 August 1998, was a loss of 8.87 standard deviations under the model, and the model gave the cumulative loss a probability of less than one in a million. Yet the model's own estimation window contained a day, 27 October 1997, on which the same positions would have lost 11.7%. The warning was in the data; the normal distribution discarded it.

The better models do better, but none of the pre-registered ones passes. A GARCH model with Student-t errors and filtered historical simulation comes closest, with an exceedance rate of 1.23% and no evidence of clustering, but it is rejected by the pre-registered one-sided test ($p = 0.029$); a two-sided test would not have rejected it ($p = 0.053$). A GARCH model with an extreme-value tail, added after the results were seen, passes both tests of its 99% VaR, although it is rejected at the 95% level and it too understates the size of the losses beyond its VaR. Correlations between the trades rose sharply in 1998, from 0.05 to 0.42 on average, but not in 2008 or 2020, when they were already moderately positive beforehand (about 0.2 to 0.3). Finally, the synthetic portfolio fails the pre-registered validation check: it lost money in August and September 1998, as LTCM did, but not in May and June. The trades rebuilt here capture the crash, not the whole of LTCM's book, and the results should be read with that limit in mind.

The remainder of the paper is organised as follows. Chapter 2 reviews the literature on LTCM, on the limits of arbitrage, on fat tails and risk models, and on crisis correlations. Chapter 3 sets out the pre-registered design, the data, the construction of the trades and the tests. Chapter 4 presents the results, and Chapter 5 discusses what they imply for the question in the title and for risk management more generally. Chapter 6 concludes.

2. Literature Review

This chapter reviews the literature on which the study builds. It begins with the history of LTCM and the explanations that have been offered for its failure, turns to the theory of arbitrage under constraints and to the empirical evidence on the trades LTCM held, and then reviews the evidence on fat tails and the risk models and backtests that were developed in response. It closes with the literature on correlations in crises and identifies the gap that the present study addresses.

2.1 Long-Term Capital Management: strategy, risk model and collapse

LTCM was founded in 1994 by John Meriwether, the former head of bond arbitrage at Salomon Brothers, with a group of partners that included Robert Merton and Myron Scholes. Its core business was relative-value or convergence trading: buying a cheaper asset and selling a closely related but more expensive one, and waiting for the price gap to narrow (Lowenstein, 2000). The best known example was the spread between the most recently issued 30-year Treasury bond and the bond issued six months earlier, which traded a few basis points cheaper despite almost identical cash flows. Other positions included Italian government bonds against interest rate swaps, mortgage securities hedged with Treasuries, swap spreads in several currencies and, from 1997, equity pairs, merger arbitrage and the sale of long-dated equity index options.

Because each spread was small, the positions were leveraged heavily. The President's Working Group on Financial Markets (1999) reports that at the start of 1998 LTCM held balance-sheet assets of more than 25 times its capital, together with derivative positions whose notional value exceeded a trillion dollars. Lowenstein (2000) describes how the fund obtained repurchase financing with almost no haircut and derivatives with no initial margin, which meant that the positions required very little capital. Risk was controlled by targeting a level of volatility, roughly that of the US stock market, estimated from historical data and combined across positions on the assumption that the trades were largely unrelated.

The losses began in May and June 1998, when LTCM lost 6.7% and 10% of its capital, as other arbitrage desks, notably at Salomon Smith Barney, reduced similar positions (Lowenstein, 2000). After Russia's default on 17 August, credit spreads widened across the world, investors sought the most liquid government bonds, and implied volatility on equity indices rose sharply. LTCM lost 44% of its capital in August. In September, as counterparties who knew its positions traded against them and its capital fell to little more than a billion dollars against assets of more than $100 billion, the fund could no longer meet its obligations, and on 23 September a consortium of fourteen banks injected $3.65 billion in exchange for 90% of the fund.

The explanations offered since fall into three groups. Jorion (2000) argues that LTCM's risk management underestimated risk because it relied on short and calm samples, assumed stable correlations and gave too little weight to stress scenarios and to the liquidity of its

positions. Edwards (1999) focuses on leverage and on the systemic risk that the rescue was intended to contain. MacKenzie (2003) offers a sociological account: LTCM's success led other arbitrageurs to copy its trades, creating a "superportfolio" of overlapping positions held by many institutions, so that when some were forced to sell, the correlations between trades that were economically unrelated rose sharply. The present study takes up the first of these explanations and asks how much of the failure a better risk model could have prevented.

2.2 The limits of arbitrage

Shleifer and Vishny (1997) provide the theoretical framework for understanding why convergence trades can fail even when they are right in the long run. In their model, arbitrage is carried out by specialised managers who invest other people's money. When prices move against them, investors judge the manager by recent performance and withdraw capital, so the arbitrageur is forced to reduce positions precisely when the mispricing, and therefore the expected return, is greatest. Arbitrage is therefore limited, and extreme mispricing can persist or worsen. The authors discuss the risk that such a manager faces when many arbitrageurs hold similar positions, a mechanism that the LTCM episode illustrated a year later.

Brunnermeier and Pedersen (2009) formalise the link between the ease of trading an asset and the ease of financing it. When volatility rises, lenders demand larger margins, which forces leveraged traders to sell, which reduces market liquidity and raises volatility further. These "liquidity spirals" explain why liquidity can dry up suddenly, why it moves together across assets, and why assets that are hard to trade suffer most in a flight to quality. Mitchell and Pulvino (2012) document such a spiral in 2008, when the withdrawal of financing by prime brokers pushed the prices of closely related securities, such as convertible bonds and their underlying shares, far apart, and show that new capital took months to arrive. Schrimpf et al. (2020) describe a similar mechanism in March 2020, when leveraged relative-value positions in US Treasuries were unwound as margins rose.

2.3 Evidence on LTCM-style trades

Several of LTCM's trades have been studied directly. Duarte et al. (2007) examine the returns to five fixed-income arbitrage strategies, including swap spread and yield curve arbitrage. They find that strategies requiring more intellectual capital to implement produce positive excess returns with a limited exposure to market risk, while simpler strategies such as swap spread arbitrage earn little and are exposed to rare large losses. The subtitle of their paper, which asks whether such strategies amount to picking up nickels in front of a steamroller, gives the present study its title. Krishnamurthy (2002) studies the spread between on-the-run and off-the-run 30-year Treasury bonds, LTCM's signature trade. He finds that the spread reflects the premium investors pay for the most liquid bond, that it widens when demand for liquid assets rises, and that the profits of the convergence trade are largely offset by the cost of borrowing the on-the-run bond in the repo market.

Selling equity volatility is the purest example of a strategy that earns a steady premium in exchange for exposure to rare large losses. Carr and Wu (2009) show that the variance risk premium, the difference between implied and subsequently realised variance, is strongly negative for the US stock index: sellers of variance swaps have earned large average returns, but the returns are highly skewed and concentrated losses occur exactly when the stock market falls. LTCM sold long-dated index volatility in 1997 and 1998 at implied levels of around 19% and lost about $1.3 billion on the position as implied volatility rose (Lowenstein, 2000).

2.4 Fat tails, volatility clustering and value-at-risk

The statistical assumptions behind LTCM's risk model were challenged long before 1998. Mandelbrot (1963) showed that changes in cotton prices had far more extreme values than a normal distribution allows and proposed stable distributions with infinite variance to describe them. Fama (1965) confirmed that the daily returns of the Dow Jones stocks were leptokurtic: large moves were many times more frequent than the normal model predicts. Under a normal distribution, a daily fall of five standard deviations should occur about once every 14,000 years; in stock returns such falls occur every few years.

A second stylised fact is that volatility clusters. Bollerslev (1986) introduced the generalised autoregressive conditional heteroskedasticity (GARCH) model, in which today's variance depends on yesterday's squared return and yesterday's variance. A model estimated with equal weights over five years adjusts only slowly when volatility rises, whereas a GARCH model reacts within days. J.P. Morgan and Reuters (1996) built the RiskMetrics methodology on a simpler exponentially weighted moving average of squared returns with a decay factor of 0.94 for daily data, which made fast-updating volatility the industry standard in the late 1990s.

Two further families of models address fat tails directly. Historical simulation takes the empirical distribution of past portfolio returns as the forecast, making no assumption about its shape but reacting slowly to changes in volatility. Barone-Adesi et al. (1999) proposed filtered historical simulation, in which past returns are first standardised by their GARCH volatility and then rescaled by the current volatility forecast. McNeil and Frey (2000) combined a GARCH filter with extreme value theory, fitting a generalised Pareto distribution to the tail of standardised losses, and showed that the combination produced more accurate estimates of extreme quantiles than either component alone. These models form the alternatives against which the LTCM-style model is compared in this study.

2.5 Backtesting risk models

A VaR forecast at the 99% level makes a testable promise: the loss should exceed it on 1% of days, and the days of exceedance should not bunch together. Kupiec (1995) proposed the proportion-of-failures test, a likelihood ratio test of whether the observed exceedance rate equals the promised one, and showed that the test has low power in samples of a few hundred days, which is why a long backtest is needed. Christoffersen (1998) added a test of

independence, which asks whether an exceedance today makes an exceedance tomorrow more likely, and combined the two into a test of conditional coverage. Because VaR says nothing about the size of losses beyond it, Acerbi and Szekely (2014) proposed backtests of expected shortfall, the average loss on the worst days, based on the realised losses on exceedance days compared with the model's own forecast.

2.6 Correlations in crises

A recurring explanation for LTCM's failure is that "correlations went to one" (Lowenstein, 2000). Longin and Solnik (2001) use extreme value theory to show that correlations between international equity markets rise in bear markets but not in bull markets, so that diversification fails when it is needed most. Forbes and Rigobon (2002) warn, however, that the simple correlation coefficient rises mechanically when volatility rises, even if the underlying relationship between two markets is unchanged. After adjusting for this bias, they find little evidence of contagion in several major crises: markets were as interdependent before the crises as during them. Any test of whether correlations rose in 1998 must therefore be interpreted with this caveat.

2.7 Research gap and positioning of the present study

The literature leaves three gaps. First, the accounts of LTCM's risk model are largely qualitative: they describe its assumptions but do not test a model of that kind against a long record of forecasts. Second, the empirical literature on arbitrage strategies has studied the returns of the trades, but rarely asks how well standard risk models would have forecast their losses, and how that performance compares across 1998, 2008 and 2020. Third, most published evaluations of risk models report a small number of preferred specifications. The present study addresses these gaps with a pre-registered, 30-year daily backtest of five risk models on synthetic LTCM-style trades built entirely from free data, with every specification reported.

3. Methodology

This chapter describes the research design. It presents the pre-registered hypotheses and decision rules, the data, the construction of the synthetic trades and the portfolio, the five risk models, the statistical tests, the power analysis and the robustness grid.

3.1 Research design and pre-registration

The study was designed as a pre-registered test. Before any result was computed, a written plan fixed the research question, the five hypotheses and their pass conditions, the data series, the formulas for the trades and the portfolio, the risk models, the statistical tests and their significance levels, the power of the primary test, the treatment of multiple testing and the robustness grid. Any departure from the plan is recorded in a deviation log, reproduced in Appendix A together with whether results had been seen at the time of the change. Not every choice was settled before results were seen: the random seed was changed, the label for H3 was decided and the treatment of the 30-year Treasury series was corrected after results had been computed. Appendix A reports each of these and, where it matters, the results under both choices. The plan was not lodged with a public registry, so the order of events cannot be verified independently. Table 1 lists the hypotheses.

Table 1. Pre-registered hypotheses and pass conditions.

#	Hypothesis	Period	Pass condition
H1	The LTCM-style model grossly understated the 1998 risk	17 Aug – 30 Sep 1998	Worst daily loss ≥ 5 model standard deviations and model probability of the cumulative loss < 1 in 10,000
H2 (primary)	The LTCM-style 99% one-day VaR fails its backtest	Jan 1995 – May 2025	One-sided Kupiec test rejects at 5%
H3	A fat-tailed, volatility-updating model passes where the LTCM-style model fails	Jan 1995 – May 2025	GARCH-t filtered HS not rejected by the Kupiec or Christoffersen test at 5%
H4	Correlations between the trades jump in crises	Aug–Oct 1998; secondary: Sep–Dec 2008, Mar–Apr 2020	Average pairwise correlation above its level over the previous 5 years; bootstrap $p < 0.05$
H5 (validation)	The synthetic portfolio behaves like LTCM in 1998	May – Sep 1998	Losses in May, June, August and September, the four months in which LTCM lost money

Note: H2 is the single primary test. Within H3 the four alternative models are corrected with the Holm (1979) procedure; within H4, 2008 and 2020 are corrected together. Source: Pre-registration plan.

H2 is the single primary test. A single crisis, however dramatic, cannot carry a statistical test on its own, so the 1998 window is examined through H1 with a descriptive pass condition, while the inference rests on the 30-year backtest. H5 is a check on the design rather than a finding: if the synthetic trades do not lose money when LTCM did, they are a poor stand-in for its positions, and the results must be read accordingly. Each hypothesis is labelled Supported if it meets its pass condition, Contradicted if the evidence points significantly the other way (for

H2, if the 95% confidence interval for the exceedance rate includes 1% and excludes 1.5%; for H4, if crisis correlations are significantly below normal), and Inconclusive otherwise.

3.2 Data sources and sample

All data are free daily series from the Federal Reserve Economic Data (FRED) database of the Federal Reserve Bank of St. Louis, the Board of Governors of the Federal Reserve System and the Kenneth French Data Library (French, n.d.). Table 2 lists the series. The files were downloaded on 16 and 17 June 2025 (Appendix D). The sample ends on 31 May 2025, and no observation after that date is used in any calculation.

Table 2. Data series and coverage.

Series	Source	Used for	Coverage
VIXCLS	FRED (Cboe)	Implied volatility sold by T1	Jan 1990 on
Market excess return + risk-free rate, daily	Kenneth French Data Library	Realised variance in T1	1926 on
DBAA	FRED (Moody's)	Baa corporate bond yield in T2	1986 on
DGS10, DGS30	FRED (US Treasury)	On-the-run Treasury yields in T2–T4	DGS10 1962 on; DGS30 1977 on, Feb 2002 – Feb 2006 excluded (no 30-year issuance)
SVENPY10, SVENPY30	Federal Reserve (Gürkaynak et al., 2007)	Off-the-run fitted par yields in T3	1971 on for 10 years
DSWP10	FRED (Federal Reserve H.15)	10-year swap rate in T4	Jul 2000 – Oct 2016

Note: All series are daily. Every calculation uses data up to 31 May 2025 only. Yields are converted to decimals. Source: Author's own compilation.

The binding constraint is the VIX, which begins in January 1990. Because every risk model is estimated on the previous five years of data, the first forecast is made in January 1995, and the backtest runs from 3 January 1995 to 30 May 2025. Two proxies require comment. The on-the-run Treasury yield is taken from the constant-maturity series, which is based on the most recently auctioned securities, while the off-the-run yield is taken from the fitted par yield curve of Gürkaynak et al. (2007), which is estimated from off-the-run securities and excludes the most recently issued bonds. The difference between the two therefore measures the premium for the newest issue, the object of LTCM's first trade, although it also contains some noise from the curve-fitting. The trade uses the 10-year maturity rather than LTCM's 30-year, because the US Treasury issued no new 30-year bonds between late 2001 and early 2006, so the 30-year constant-maturity yield does not measure a genuine on-the-run bond in those years. FRED's notes describe the series as discontinued from 18 February 2002 to 9 February 2006, yet the file downloaded from FRED contains values for this period. Their origin is not documented in the

file; they appear to be estimates rather than yields on a traded 30-year bond, since none was issued. As the plan specified, the 30-year version of the trade, which enters only the robustness grid, therefore excludes the days from 18 February 2002 to 9 February 2006. The 10-year swap rate is available only from July 2000 to October 2016 (its last observation is 28 October 2016), so the swap-spread trade, LTCM's largest single loss, cannot be tested for 1998.

3.3 Construction of the synthetic trades

Each trade is expressed as a daily profit and loss per unit of exposure and is designed to have the defining feature of LTCM's positions: a small, steady income in normal times and large losses when investors seek safety and liquidity.

T1: short equity volatility. The first trade sells a 30-day variance swap on the US stock market at the level of the VIX and rolls it daily. Each day, one twenty-first of the contract settles against that day's realised variance, and the remaining twenty-twenty-firsts are marked to the new level of implied variance:

$$P_{volt} = (1/21)\,(\mathrm{VIX}^2_{t-1} - 252\,r^2_t) - (20/21)\,(\mathrm{VIX}^2_{t} - \mathrm{VIX}^2_{t-1})$$

where VIX_t is the VIX in percentage points and r_t is the daily total return of the US stock market in percent, taken from the Kenneth French Data Library as the market excess return plus the risk-free rate. The profit is then divided by $2 \times \mathrm{VIX}_{t-1}$ to express it per point of volatility, the unit in which LTCM measured its option exposure.

T2: credit convergence. The second trade holds Moody's seasoned Baa corporate bonds against the 10-year Treasury note with matched duration. It earns the yield spread as carry and loses when the spread widens:

$$P_t = s_{t-1} / 252 - D\,(s_t - s_{t-1})$$

where s_t is the spread in decimal form, here Baa minus the 10-year Treasury yield, and D is the assumed duration, set to 10 for the credit trade.

T3: liquidity convergence. The third trade holds the off-the-run 10-year Treasury, measured by the fitted par yield, against the on-the-run 10-year note, measured by the constant-maturity yield. It follows the same formula with s_t equal to the fitted par yield minus the constant-maturity yield and a duration of 8.

T4: swap spread. The fourth trade receives the fixed rate on a 10-year interest rate swap and sells the 10-year Treasury note, so that it gains when the swap spread narrows. Its daily profit is −8 times the change in the spread, with carry ignored. Because the swap rate is available only from July 2000, T4 enters only the portfolio used for the later crises.

On the few days when the stock market was open but the bond market was closed, the bond spreads are carried forward from the previous day, for at most three days. This implementation detail is recorded in the deviation log. Spreads are not carried forward past the end of a discontinued series.

3.4 Portfolio construction

The three trades are combined into a single portfolio, rebalanced on the first trading day of each month using only data available up to the previous day. First, each trade receives a weight inversely proportional to its volatility over the previous five years (1,260 trading days), so that the three trades contribute equal stand-alone risk. Second, the weights are scaled so that the predicted volatility of the portfolio, using the five-year covariance matrix, is 15% a year:

$$w_{i,t} = k_t / \hat{\sigma_{i,t}}, \quad \text{with } k_t \text{ chosen so that } \sqrt{(252\ w_t'\ \Sigma_t\ w_t)} = 15\%$$

where Σ_t is the five-year covariance matrix of the trades and w_t the vector of weights. The daily return of the portfolio, as a fraction of capital, is $R_t = w_t'\ P_t$, where P_t is the vector of the three trades' daily profits. Leverage is therefore not chosen but results from the volatility target. It is reported in Chapter 4 and compared with LTCM's. The target does not affect the backtests, because every model's forecast scales in proportion to the weights; it matters only for the size of losses and for leverage, which are reported for targets of 10%, 15% and 20%.

3.5 Risk models

Each day, five models forecast the portfolio's one-day VaR at the 99% and 95% levels and its expected shortfall at 97.5%. Every model uses only the previous 1,260 days of the trades' profits, combined with the weights actually held that day, so that its forecast is fully out of sample. Table 3 summarises the models.

Table 3. The risk models.

Model	Distribution and volatility	Why it is included
M1 Normal, 5-year	Normal; volatilities and correlations from the previous 5 years, equal weights, zero mean	Mirrors the approach described by LTCM (the model of interest)
M2 EWMA	Normal; exponentially weighted covariance, decay 0.94	Industry standard of the late 1990s (J.P. Morgan & Reuters, 1996)
M3 Historical simulation	Empirical distribution of the previous 1,260 portfolio returns	No distributional assumption, but slow to react
M4 GARCH-t filtered HS	GARCH(1,1) with Student-t errors; empirical standardised residuals	Fat tails and fast-updating volatility (H3)
M5 Extreme value	Generalised Pareto distribution fitted to raw losses beyond their 95th percentile (unconditional)	Designed for rare, extreme losses
M6 GARCH-EVT (post hoc)	GARCH(1,1)-t filter as in M4; generalised Pareto distribution fitted to the largest 10% of standardised losses (McNeil and Frey, 2000)	Supplementary; added after the results were seen, primary specification only

Note: All models use the previous 1,260 trading days and the portfolio weights held on the forecast day, and are re-estimated daily. M1–M5 are pre-registered. At the 95% level M5 equals M3 by construction, because its threshold is the 95th percentile. Source: Pre-registration plan; M6 added by the author.

M1 is the model of interest. It assumes that the portfolio return is normally distributed with a mean of zero and a standard deviation equal to the sample standard deviation over the previous five years, so that its 99% VaR is

$$\mathrm{VaR}_t = z_{0.99} \times \hat{\sigma_t}$$

where $z_{0.99} = 2.326$ is the 99% quantile of the standard normal distribution and $\hat{\sigma_t}$ the five-year standard deviation. M2 replaces the equally weighted standard deviation with the RiskMetrics exponentially weighted covariance matrix with a decay factor of 0.94, updated every day. M3 takes the empirical quantile of the previous 1,260 portfolio returns. M4 fits a GARCH(1,1) model with Student-t errors to the previous 1,260 returns every day, standardises the returns by their fitted volatility, and rescales the empirical quantile of the standardised returns by the next day's volatility forecast (Barone-Adesi et al., 1999). M5 fits a generalised Pareto distribution to the raw portfolio losses above their 95th percentile and reads the VaR from the fitted tail. This is the unconditional peaks-over-threshold method, not the conditional model of McNeil and Frey (2000), who fit the tail to GARCH-filtered residuals; because it ignores the current level of volatility, it reacts as slowly as historical simulation. Its threshold also means that its 95% VaR is, by construction, the empirical 95th percentile, so at the 95% level M5 is identical to M3 and is not counted separately. All models are re-estimated every day. Because the conditional model of McNeil and Frey (2000) is a standard benchmark, it was added after the results were seen as a supplementary model, M6, in the primary specification only: a GARCH(1,1)-t filter as in M4, with a generalised Pareto distribution fitted to the largest 10% of the standardised losses.

3.6 Statistical tests and decision rules

An exceedance occurs on day t when the loss is larger than the VaR forecast made for that day, that is when $R_t < -\mathrm{VaR}_t$. If the model is correct, the number of exceedances x in n days follows a binomial distribution with probability p = 1%. The Kupiec (1995) test compares the likelihood of the observed exceedance rate with that of the promised rate:

$$\mathrm{LR}_{\mathrm{POF}} = -2 \ln[(1-p)^{n-x} p^{x}] + 2 \ln[(1 - x/n)^{n-x} (x/n)^{x}]$$

which is distributed as a chi-squared variable with one degree of freedom under the null. Because the hypotheses concern models that underestimate risk, the pre-registered test is one-sided, with the exact binomial probability of observing x or more exceedances, and this test is used for every rejection for too many exceedances reported in the paper, including the robustness grid; only the finding that a model has too few exceedances relies on the two-sided test. The two-sided likelihood ratio p-value and an exact 95% confidence interval for the exceedance rate are also reported; where the two tests disagree, the text says so. The Christoffersen (1998) test compares the probability of an exceedance following an exceedance with that following a normal day, and the Acerbi and Szekely (2014) Z2 statistic compares the

realised losses on the worst 2.5% of days with each model's expected shortfall, with its p-value obtained from 1,000 simulated paths drawn from each model's own daily forecast distribution. A negative Z2 indicates that the model underestimates the size of tail losses.

For the 1998 window (H1), two measures are computed for every model: the loss on the worst day of the window divided by the model's standard deviation for that day, using its forecast for that day made at the previous close, and the probability that the model, as estimated at the close of 14 August 1998, assigns to a cumulative loss at least as large as the one observed over the 32 days from 17 August to 30 September. For the normal models, the cumulative probability follows from scaling the daily standard deviation by the square root of the number of days; for the other models, it is computed by summing 32 independent draws from the model's daily distribution, two million times. For M4 this holds volatility at its 14 August forecast and so ignores the GARCH dynamics, which raise volatility after large losses and fatten the multi-day tail; a version that updates volatility along each simulated path is reported alongside as a post-hoc check.

For H4, the average pairwise correlation of the three trades' daily profits in each crisis window is compared with the average over the previous 1,260 days. Inference uses the stationary bootstrap of Politis and Romano (1994), with a mean block length of five days and 10,000 draws, which preserves the short-run dependence in the data. The crisis windows are 1 August to 31 October 1998, 15 September to 31 December 2008 and 1 March to 15 April 2020.

H2 is the single primary test, evaluated at the 5% level. Within H3, the four alternative models are corrected for multiple testing with the Holm (1979) step-down procedure. Within H4, 1998 is the lead test at the 5% level, and 2008 and 2020 are corrected together with the Holm procedure. Every model, trade and crisis window is reported whatever the outcome.

3.7 Power analysis

Table 4 reports the power of the primary test. The backtest covers about 7,700 trading days, so a correct 99% VaR should be exceeded about 77 times, and a one-sided test at the 5% level rejects when there are about 92 or more exceedances. The test has about 99% power to detect a model whose true exceedance rate is 1.5% instead of 1%, and about two chances in three of detecting a rate of 1.25%. A non-rejection would therefore be informative about large failures but not about small ones.

Table 4. Power of the primary test (one-sided Kupiec test at 5%, about 7,700 days).

True exceedance rate	Expected exceedances	Power
1.00% (correct model)	77	5% (size)
1.25%	96	about 66%
1.50%	116	about 99%
2.00% or more	154 or more	about 100%

Note: The test rejects when there are about 92 or more exceedances. Power computed from the binomial distribution before any results were seen. Source: Pre-registration plan.

3.8 Robustness grid

The plan fixed a grid of alternative specifications before any results were seen. The estimation window is two years, five years or expanding from the start of the data; the durations of the credit and liquidity trades are 7 and 6, 10 and 8, or 13 and 10 years; the liquidity trade uses the 10-year or the 30-year maturity; and the VaR level is 99% or 95%. Every combination is run for all five models, giving 36 specifications per model and 180 in total, and all are reported. One further portfolio adds the swap-spread trade (T4) and is backtested from July 2005, when T4 first has five years of history, to October 2016, so that LTCM's largest trade can enter at least the 2008 test.

4. Results

This chapter presents the results. It first describes the synthetic trades and the portfolio and checks whether they behaved like LTCM in 1998. It then reports how the risk models saw the 1998 window, the 30-year backtests, the correlations in crises, leverage and the 2008 swap test, and the robustness grid.

4.1 Descriptive statistics

Table 5 summarises the four trades over the backtest period. All four show the pattern the design intended. The short-volatility trade earns money on two days in three and has an annualised Sharpe ratio of 0.64, but its daily returns have a skewness of −5.4 and an excess kurtosis of 68: most of the time it collects a small premium, and occasionally it loses many months of income in a day. The credit trade has a Sharpe ratio of 0.45 and a skewness of −1.8. The liquidity trade earns very little, with a Sharpe ratio of 0.09, which is consistent with the finding of Krishnamurthy (2002) that the convergence trade earns little. The trade here also ignores the cost of borrowing the on-the-run bond in the repo market, which Krishnamurthy shows absorbs most of the profit, so its true return would be lower still. Its excess kurtosis of 63 shows that its rare large moves dominate its risk.

Table 5. Descriptive statistics of the synthetic trades and the portfolio (daily returns).

Trade	Period	Days	Sharpe ratio	Skew	Excess kurtosis	Days > 0 (%)	Worst day
T1 Short volatility	Jan 1995 – May 2025	7,654	0.64	−5.42	67.7	65.9	16 Mar 2020
T2 Credit convergence	Jan 1995 – May 2025	7,654	0.45	−1.78	21.5	60.9	17 Sep 2001
T3 Liquidity convergence	Jan 1995 – May 2025	7,654	0.09	−0.58	63.4	51.9	14 Nov 1996
T4 Swap spread	Jul 2000 – Oct 2016	4,107	0.10	0.03	7.6	45.9	18 Mar 2009
Portfolio (T1–T3, 15% target)	**Jan 1995 – May 2025**	**7,654**	**0.53**	**−3.14**	**31.8**	**–**	**09 Mar 2020**

Note: Sharpe ratios are annualised (mean × 252 divided by standard deviation × √252). Trades are measured per unit of exposure, so only unit-free statistics are shown. The portfolio earned 8.8% a year with a volatility of 16.8%. Source: Author's own calculations.

The combined portfolio, rebalanced monthly to a 15% volatility target, earned 8.8% a year with a realised volatility of 16.8% and a Sharpe ratio of 0.53 (computed before rounding). Its daily returns have a skewness of −3.1 and an excess kurtosis of 32. Its five worst days were 9 and 12 March 2020 (−15.1% and −14.9%), 27 October 1997 (−14.4% with the weights then held), 5 February 2018 (−12.7%) and 15 October 2008 (−12.1%). Measured by simple sums of daily returns, the portfolio lost 96% of its capital in 2008 and more than 20% in 2007, 2018 and 2020. Summing returns holds capital constant, which implicitly refills the portfolio after every loss. With compounding, the same daily returns give a loss of 64% in 2008 and of 71% over

2007 and 2008 together, and a peak-to-trough drawdown of 76% from February 2007 to December 2008. The 15%-target portfolio is never wiped out, but no fund that lost three quarters of its capital would keep its investors or its lenders, so the constant-capital figures overstate what a real fund could have survived. August 1998, at −18.0%, is only the fourth worst month in the sample, after October 2008, March 2020 and September 2008. The 1998 crisis was severe for these trades, but it was not the worst they would face.

The investor letter of 1994 quoted by Lowenstein (2000) gives a rough benchmark. In Lowenstein's quotation, the letter stated that "roughly, over a long period of time, investors may experience a loss of 5% or more in about one month in five, and a loss of 10% or more in about one month in ten"; Lowenstein adds that "only one year in fifty should it lose at least 20 percent of its portfolio" (Lowenstein, 2000, p. 63). Read literally, these figures cannot all come from one normal distribution: a 10% monthly loss one month in ten implies an annual volatility of about 27%, at which a 20% annual loss would occur about one year in five, not one in fifty. Read as annual frequencies, they fit a single normal distribution with an expected annual return of about 5% and a volatility of about 12%, so the letter may well have meant years, although Lowenstein's text says months. Under the monthly reading, 33 of 365 months (9.0%) of the synthetic portfolio lost 5% or more and 14 (3.8%) lost 10% or more, fewer than the letter implied. Under the annual reading, of the 30 complete calendar years from 1995 to 2024, 6 (20%) lost 5% or more, 5 (17%) lost 10% or more and 4 (13%) lost 20% or more, against the letter's 20%, 10% and 2%. Under either reading, the frequency of severe annual losses, four in 30 years instead of one in fifty, is the figure on which the letter was most wrong.

4.2 Validation: did the portfolio behave like LTCM in 1998?

Table 6 compares the monthly returns of the synthetic portfolio in 1998 with those reported for LTCM. The portfolio lost 18.0% in August and 10.8% in September, the two months in which LTCM's losses became fatal. It did not lose money in May and June, however, when LTCM lost 6.7% and 10% as other arbitrage desks reduced their positions. The pre-registered validation condition required losses in all four months; the portfolio loses in two, and H5 is therefore not met.

Table 6. Monthly returns in 1998: synthetic portfolio and LTCM (H5).

Month	Synthetic portfolio (%)	LTCM reported (%)	Both lose?
May 1998	3.81	−6.7	No
June 1998	3.47	−10.0	No
July 1998	−0.35	about 0	–
August 1998	−18.04	−44.0	Yes
September 1998	−10.76	most of the remaining capital	Yes

Note: Portfolio returns are sums of daily returns at the 15% volatility target. LTCM figures from Lowenstein (2000); September is approximate. The pass condition requires losses in May, June, August and September: two of four are met, so H5 is not supported. Source: Author's own calculations.

The failure is informative. LTCM's losses in May and June came from positions this study cannot rebuild with free data, such as swap spreads, mortgage securities and equity pairs, and from the selling of arbitrage positions by other institutions, which affected LTCM's book directly but did not move the broad credit, liquidity and volatility measures used here. What the synthetic portfolio captures is the market-wide flight to quality and liquidity that followed the Russian default. The results below are therefore about how risk models would have seen that common crash in LTCM-style exposures, not about LTCM's full book.

4.3 The 1998 crisis through the eyes of the risk models

Figure 1 shows the daily returns of the portfolio from June to October 1998, together with the 99% VaR of the LTCM-style model (M1) and of the GARCH-t model (M4). Before August, the two forecasts are almost identical. After the Russian default, M1's forecast barely moves, because one crisis day adds little to a five-year average, whereas M4's forecast rises at once and roughly doubles after the losses of late August. The portfolio lost more than M1's VaR on 8 of the 32 days from 17 August to 30 September, where a correct model would be exceeded less than once.

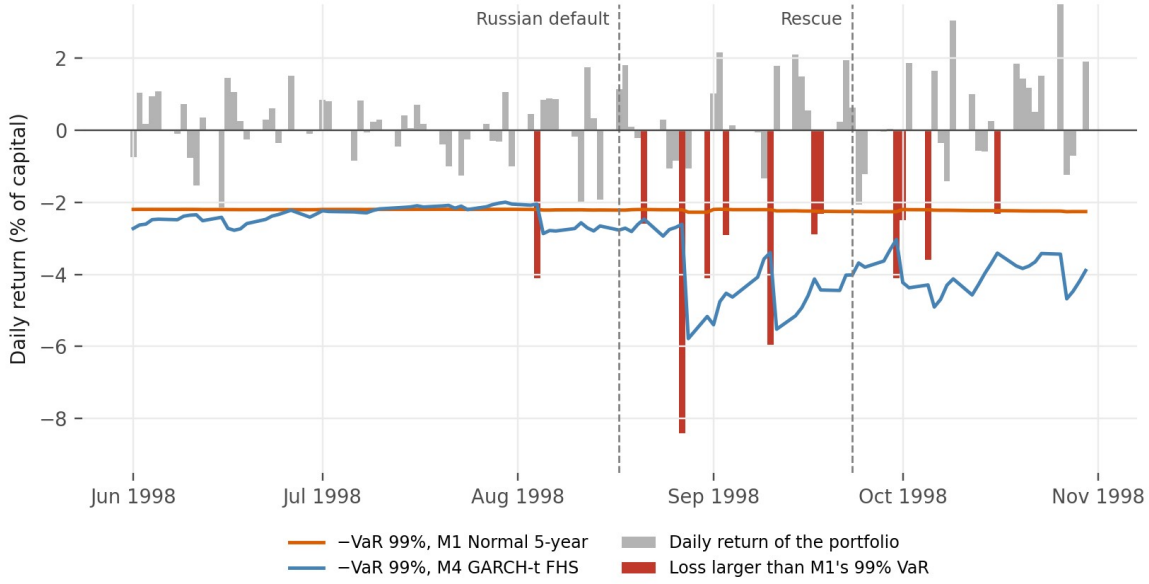


Figure 1. Daily returns of the synthetic portfolio and the 99% value-at-risk of the LTCM-style model (M1) and the GARCH-t filtered historical simulation model (M4), June to October 1998. Red bars mark days on which the loss exceeded M1's forecast. Figure produced using Matplotlib (Hunter, 2007).

Table 7 reports the H1 measures for every model. Over the 32 trading days of the window the portfolio lost 25.7% of its capital. The worst day was 27 August 1998, when the portfolio lost 8.41%, the equivalent of 8.87 standard deviations under the LTCM-style model. Under a normal distribution, a daily loss of this size has a probability of about 4 in 10^{19}. The model gave the cumulative loss over the window a probability of 8.7×10^{-7}, or about one in 1.1 million. Both pre-registered conditions are met by a wide margin, and H1 is Supported.

Table 7. The 17 August – 30 September 1998 window through each model (H1).

Model	VaR 99% on 27 Aug (%)	Worst day in model SDs	P(cumulative loss ≥ 25.7%)	Exceedances in 32 days	H1 conditions met
M1 Normal, 5-year	2.21	8.87	8.7×10^{-7}	8	Both
M2 EWMA (RiskMetrics)	2.95	6.63	8.7×10^{-5}	2	Both
M3 Historical simulation	2.48	8.87	6.0×10^{-5}	6	Both
M4 GARCH-t filtered HS	2.61	7.92	2.4×10^{-4}	4	SD only
M5 Extreme value (GPD)	2.58	8.87	7.3×10^{-4}	7	SD only

Note: The VaR and the worst-day measure use each model's forecast for 27 August 1998, made at the close of 26 August; the cumulative probability uses the forecast made at the close of 14 August. Worst day: 27 August 1998, loss of 8.41%. Cumulative loss over 32 trading days: 25.73%. M3 and M5 use the sample standard deviation for the SD measure. Probabilities for M1–M2 are analytical; for M3–M5 they come from 2,000,000 simulated 32-day paths of independent daily draws. For M4, simulating paths in which the GARCH volatility responds to each day's loss (post hoc) raises the probability to 6.4×10^{-4}. Expected exceedances in 32 days under a correct model: 0.32. H1 concerns M1. Source: Author's own calculations.

The loss was concentrated in the credit trade. Of the 25.7% cumulative loss, 17.4 percentage points came from the credit trade, 4.5 from the liquidity trade and 3.9 from the short-volatility trade, although ex ante each contributed roughly a third of the portfolio's risk (27%, 35% and 38%). The worst single day, 27 August, was the day on which credit spreads widened sharply and equity implied volatility jumped as the Russian crisis spread to other emerging markets. By contrast, the portfolio lost only 2.6% on 21 August, LTCM's worst day of the month, a 2.7 standard-deviation move: the synthetic trades do not include the swap and equity positions that drove LTCM's losses that day.

The most striking number in Table 7 is not in the 1998 window but in the model's own estimation sample. On 27 October 1997, when the Asian crisis reached the US stock market, the portfolio held in August 1998 would have lost 11.7% in a single day (the 14.4% loss reported in Section 4.1 uses the weights actually held in October 1997), more than on any day of the 1998 crisis. The LTCM-style model had that day in its five-year window. It averaged it together with 1,259 calm days into a standard deviation of 0.95% and then assigned losses of that size a probability of practically zero. The historical-simulation and extreme-value models, which keep that day as a tail observation, gave the cumulative 1998 loss probabilities between 6×10^{-5} and 7×10^{-4}: small, but not astronomical. The warning was present in the data; it was the normal distribution that discarded it.

4.4 Thirty years of backtests

Table 8 reports the backtests of the 99% VaR over the full sample of 7,654 days. The LTCM-style model is exceeded on 173 days, against 77 expected, an exceedance rate of 2.26% with a 95% confidence interval of 1.94% to 2.62%. The one-sided Kupiec test rejects with a p-value below 0.001, and the interval excludes both 1% and 1.5%. H2, the primary pre-registered

test, is Supported: the LTCM-style model fails its backtest. The failure is not only one of frequency. There are 20 pairs of consecutive exceedance days, where fewer than one would be expected under a correct model and about four if exceedances occurred independently at the observed rate of 2.26%; the Christoffersen test, which conditions on the observed rate, rejects independence, and on days when M1 is exceeded the average loss is 1.8 times the VaR. The Z2 statistic of −0.99 indicates that the model's expected shortfall understates the realised tail losses by almost half.

Table 8. Backtests of the 99% one-day VaR, January 1995 – May 2025 (H2 and H3).

Model	Count	Rate (%)	95% CI (%)	Kupiec p (one-sided)	Back-to-back	Clustering p	Z2	Z2 p
M1 Normal, 5-year	173	2.26	[1.94, 2.62]	< 0.001	20	< 0.001	−0.99	< 0.001
M2 EWMA (RiskMetrics)	172	2.25	[1.93, 2.60]	< 0.001	6	0.303	−0.93	< 0.001
M3 Historical simulation	108	1.41	[1.16, 1.70]	< 0.001	10	< 0.001	−0.35	< 0.001
M4 GARCH-t filtered HS	94	1.23	[0.99, 1.50]	0.029	2	0.470	−0.20	0.015
M5 Extreme value (GPD)	99	1.29	[1.05, 1.57]	0.007	8	< 0.001	−0.31	0.001
M6 GARCH-EVT (post hoc)	86	1.12	[0.90, 1.39]	0.152	1	0.973	−0.18	0.021

Note: 7,654 trading days; expected exceedances at 1%: 76.5. The one-sided Kupiec p-value is the binomial probability of at least the observed number of exceedances. The 95% CI is the exact Clopper–Pearson interval. Count = number of exceedances. Back-to-back = days on which an exceedance followed an exceedance. Clustering p is the Christoffersen (1998) independence test. Z2 is the Acerbi and Szekely (2014) statistic (0 for a correct model, negative when tail losses are underestimated); its p-value is from 1,000 simulated paths per model. Holm-adjusted Kupiec p-values for M2–M5 are all below 0.05. For M4 the 95% CI includes 1% and the two-sided p-value is 0.053, so its rejection rests on the pre-registered one-sided test. M6 was added after the results were seen and is not part of any hypothesis test. Source: Author's own calculations.

The failure is not confined to crises. Of M1's 173 exceedances, 39 fall in the 172 days of the three crisis windows, but the remaining 134 occur on the other 7,482 days, a rate of 1.79%, which on its own also rejects the model ($p < 0.001$). The LTCM-style model is too optimistic in ordinary years as well.

Figure 2 shows how the exceedances accumulate over time. The two normal models (M1 and M2) drift steadily above the expected line, with jumps in 1998, 2000–2001, 2007–2008 and 2020. The three models that allow for fat tails (M3, M4 and M5) stay much closer to it, but all of them end the sample above it, mainly because of the jumps in 2008 and 2020.

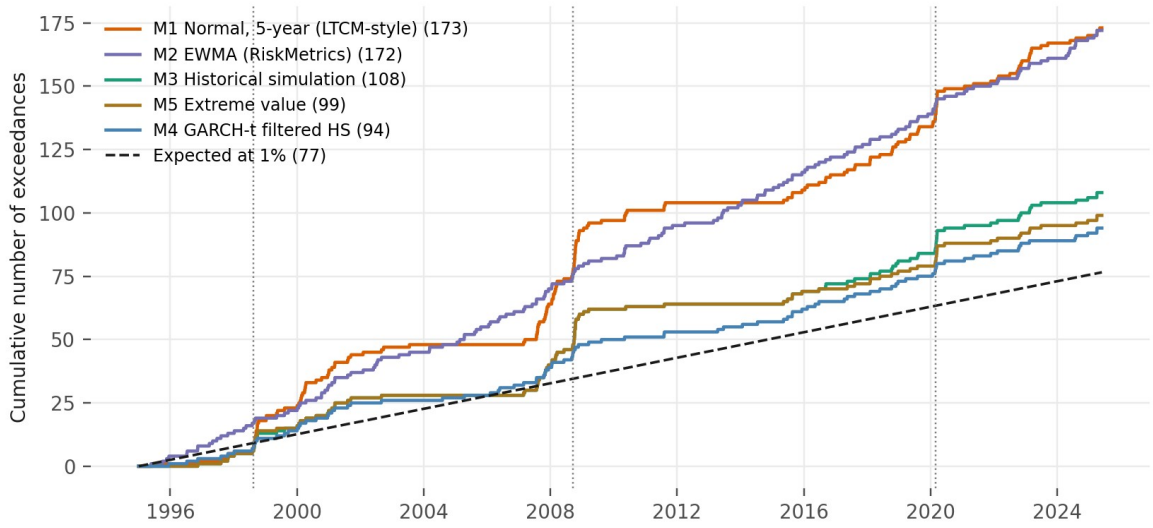


Figure 2. Cumulative number of days on which the loss of the synthetic portfolio exceeded each model's 99% value-at-risk, January 1995 to May 2025, compared with the number expected under a correct model. Dotted lines mark August 1998, September 2008 and March 2020. Total exceedances in brackets. Figure produced using Matplotlib (Hunter, 2007).

The comparison between the alternative models is instructive. The RiskMetrics model (M2) reacts quickly to volatility, so its exceedances do not cluster (Christoffersen p = 0.30), but it retains the normal distribution and is exceeded on 172 days, as often as M1. Historical simulation (M3) and extreme value theory (M5) capture the fat tails, with 108 and 99 exceedances, but react slowly, so their exceedances cluster. Only the GARCH-t filtered historical simulation model (M4) addresses both problems. Its exceedance rate of 1.23% is close to the promised 1%, and its exceedances show no clustering (p = 0.47). Its 95% confidence interval, 0.99% to 1.50%, includes 1%, however, and the verdict depends on the choice of test. The pre-registered one-sided Kupiec test rejects it with a p-value of 0.029, and it remains rejected after the Holm correction across the four alternative models. The two-sided test, with a p-value of 0.053, would not reject it. H3 required M4 to pass both tests, so under the pre-registered test its pass condition is not met; the plan did not define a "Contradicted" outcome for H3, so H3 is labelled Inconclusive. In plain terms, H3 is not supported, but narrowly, and only because the plan specified a one-sided test. The Z2 test also rejects every model, although M4's statistic of −0.20 is by far the closest to zero.

The supplementary model M6, the GARCH-filtered extreme-value model of McNeil and Frey (2000), does better still. It is exceeded on 86 days, a rate of 1.12% with a 95% confidence interval of 0.90% to 1.39%, and neither the one-sided Kupiec test (p = 0.15) nor the Christoffersen test (p = 0.97) rejects it. Its Z2 statistic of −0.18 is still significantly negative (p = 0.021), so it too understates the size of the losses beyond its VaR. M6 was added after the results were seen and was run only in the primary specification, so it is not part of any hypothesis test, but it shows that a standard model available in 2000 would have passed the 99% backtest that the LTCM-style model fails so badly. Its pass should be read with the same caution that Section 4.7 applies to the robustness grid: M6 was added after M4 had narrowly failed, with a single

threshold choice and in a single specification, and at the 95% level it is exceeded too often (5.55%, one-sided p = 0.015).

Table 9 reports the same tests for the 95% VaR, which tell a complementary story. At this level the LTCM-style model has too few exceedances, 4.43% instead of 5% (two-sided p = 0.02), and they cluster. A normal model estimated on fat-tailed data overstates moderate losses and understates extreme ones. The RiskMetrics model passes both tests at 95%, with an exceedance rate of 5.07%, while M4 is exceeded slightly too often (5.64%, p = 0.01), and so is M6 (5.55%, one-sided p = 0.015). At this level M5 is identical to M3 by construction.

Table 9. Backtests of the 95% one-day VaR, January 1995 – May 2025.

Model	Count	Rate (%)	95% CI (%)	Kupiec p (two-sided)	Back-to-back	Clustering p
M1 Normal, 5-year	339	4.43	[3.98, 4.91]	0.020	45	< 0.001
M2 EWMA (RiskMetrics)	388	5.07	[4.59, 5.58]	0.782	21	0.755
M3 Historical simulation	425	5.55	[5.05, 6.09]	0.029	57	< 0.001
M4 GARCH-t filtered HS	432	5.64	[5.14, 6.18]	0.011	31	0.171
M5 Extreme value (GPD) ‡	425	5.55	[5.05, 6.09]	0.029	57	< 0.001
M6 GARCH-EVT (post hoc)	425	5.55	[5.05, 6.09]	0.029	28	0.350

Note: Expected exceedances at 5%: 383. Count, back-to-back and clustering p as in Table 8. A two-sided test is reported because the LTCM-style model has too few exceedances at this level. ‡ Identical to M3 by construction. M6 has the same count as M3 by coincidence: 96 of the 425 exceedance days of each model are not shared with the other. Source: Author's own calculations.

4.5 Correlations in crises

Table 10 and Figure 3 report the correlations between the three trades before and during each crisis. In the five years before August 1998, the trades were almost unrelated: the average pairwise correlation was 0.05, and the short-volatility trade was slightly negatively correlated with the credit trade. In August to October 1998, the average correlation rose to 0.42, and the correlation between the credit and liquidity trades reached 0.66. The increase of 0.37 is significant (bootstrap p < 0.001), and H4 is Supported for 1998. Diversification between the trades, which justified their combined leverage, disappeared exactly when it was needed.

Table 10. Average pairwise correlation of the three trades before and during each crisis (H4).

Crisis window	Days	Previous 5 years	Crisis window	Change	95% CI of change	p (one-sided)	Crisis, adjusted	p, adjusted
Aug–Oct 1998	64	0.05	0.42	0.37	[0.22, 0.54]	< 0.001	0.22	0.001
15 Sep – 31 Dec 2008	76	0.27	0.24	−0.03	[−0.20, 0.12]	0.651	0.05	1.000
1 Mar – 15 Apr 2020	32	0.21	0.18	−0.03	[−0.33, 0.23]	0.577	0.03	1.000

Note: Correlations of daily trade returns. The 95% CI and p-value are from a stationary bootstrap (Politis & Romano, 1994) with 10,000 draws and mean block length 5, resampling the crisis and pre-crisis windows independently. Correlations for

each pair of trades are shown in Figure 3. 2008 and 2020 are Holm-corrected together; neither is significant before or after correction. Adjusted = crisis correlation corrected for the rise in volatility as in Forbes and Rigobon (2002), using for each pair the larger of the two trades' variance increases; this adjustment was added after the results were seen. Source: Author's own calculations.

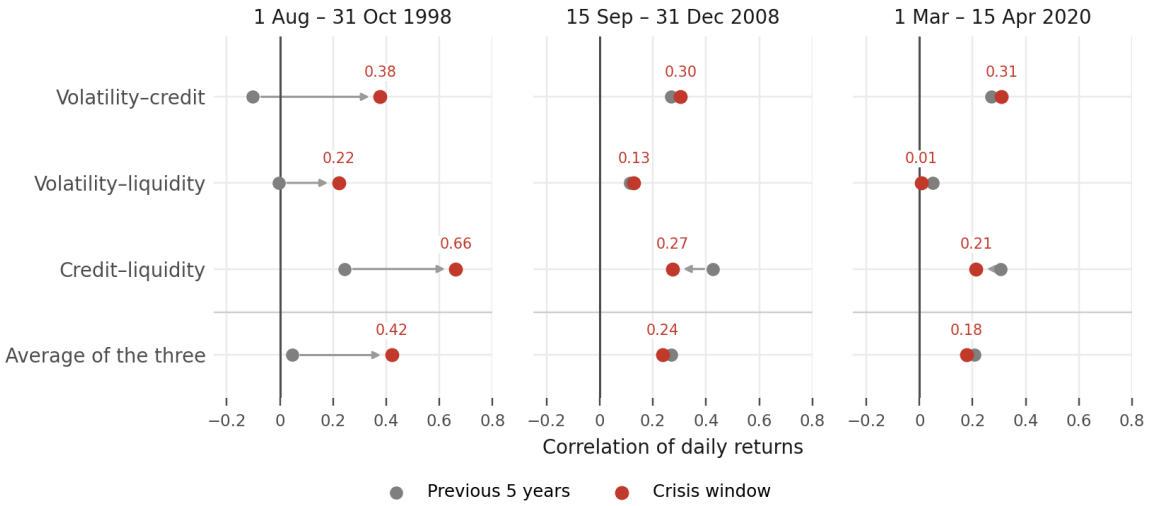


Figure 3. Pairwise correlations of the daily returns of the three synthetic trades over the previous five years (grey) and in each crisis window (red, value shown). Arrows show the direction of change. Figure produced using Matplotlib (Hunter, 2007).

The later crises tell a different story. In the last four months of 2008 and in March and early April 2020, the average correlation was slightly lower than in the previous five years (0.24 against 0.27, and 0.18 against 0.21), and neither difference is significant. One reason the correlations did not jump is that the windows before these crises already contained episodes of stress: the five years before September 2008 include the credit turmoil of 2007 and early 2008, and the five years before March 2020 include the volatility shock of February 2018. The baseline was therefore no longer calm, although at 0.21 to 0.27 it was moderate rather than high. Under the pre-registered labels, H4 is Inconclusive for 2008 and 2020.

As Forbes and Rigobon (2002) show, correlations measured in high-volatility windows are biased upward, so part of the 1998 increase may be mechanical. A post-hoc adjustment for this bias, which uses for each pair of trades the larger of the two increases in variance, lowers the 1998 crisis correlation from 0.42 to 0.22, but the increase over the pre-crisis level of 0.05 remains significant (0.18 before rounding, bootstrap p = 0.001). The H4 result for 1998 therefore survives the adjustment, although the size of the jump is roughly halved. After the same adjustment, the 2008 and 2020 crisis correlations fall to 0.05 and 0.03, below their pre-crisis levels. Even so, a correlation of about 0.25 in normal times leaves the portfolio far less diversified than the near-zero correlations that an investor in 1998 would have estimated.

4.6 Leverage and the 2008 test with the swap-spread trade

Table 11 reports the leverage implied by the volatility target and the losses it produced. At the 15% target, the portfolio held in August 1998 had long bond positions of 6.1 times

capital, or 12.2 times counting both legs, together with a short-volatility position. The loss from August to September 1998 would have been 28.8% of capital. At a 20% target, leverage rises to 16.2 times gross and the loss to 38.4%. These numbers are much smaller than LTCM's balance-sheet leverage of 25 to 30 times and its loss of more than 90%, which is consistent with the synthetic portfolio holding only three of LTCM's trade types, being unable to include the swap and equity positions that caused its largest losses, and never being forced to sell.

Table 11. Leverage and 1998 losses at different volatility targets.

Volatility target	Long bond notional (× capital)	Gross bond notional (× capital)	Loss Aug–Sep 1998 (%)	Worst day 1998 (%)
10%	4.1	8.1	19.2	5.6
15% (primary)	6.1	12.2	28.8	8.4
20%	8.1	16.2	38.4	11.2

Note: Weights held in August 1998. Bond notional is the sum of the long legs of the credit and liquidity trades; gross counts both legs. Over the whole backtest the long bond notional averaged 8.0 times capital at the 15% target. For comparison, LTCM's balance-sheet leverage was about 25–30 times and its derivative notional far larger (President's Working Group on Financial Markets, 1999). Source: Author's own calculations.

The portfolio that adds the swap-spread trade can only be tested from July 2005 to October 2016, 2,847 trading days. Table 12 reports its 99% backtests. The LTCM-style model is exceeded on 61 days against 28.5 expected, a rate of 2.14%; the Kupiec test rejects it ($p < 0.001$) and its exceedances cluster (Christoffersen $p = 0.011$). The GARCH-t model is exceeded on 35 days, a rate of 1.23%, almost exactly its rate in the main backtest, but with fewer days the test has less power and does not reject it (one-sided $p = 0.13$). In the 2008 window, from 15 September to 31 December, the four-trade portfolio lost 66.8% of its capital, and the LTCM-style model was exceeded on 15 of the 76 days. Over the same window the three-trade portfolio of the main backtest lost 84.7%, so adding the swap-spread trade, with each of four trades given an equal share of risk, reduced the 2008 loss rather than adding to it. The swap-spread trade diversified the other three in 2008, even though it was LTCM's largest loss in 1998.

Table 12. Backtests of the 99% VaR for the portfolio including the swap-spread trade (T1–T4).

Model	Count	Expected	Rate (%)	95% CI (%)	Kupiec p (one-sided)	Clustering p
M1 Normal, 5-year	61	28.5	2.14	[1.64, 2.74]	< 0.001	0.011
M2 EWMA (RiskMetrics)	68	28.5	2.39	[1.86, 3.02]	< 0.001	0.590
M3 Historical simulation	41	28.5	1.44	[1.04, 1.95]	0.015	0.142
M4 GARCH-t filtered HS	35	28.5	1.23	[0.86, 1.71]	0.129	0.452
M5 Extreme value (GPD)	39	28.5	1.37	[0.98, 1.87]	0.034	0.117

Note: Count, clustering p as in Table 8. Four trades, equal risk, 15% volatility target; backtest from 12 July 2005 to 28 October 2016, 2,847 days. From 15 September to 31 December 2008 (76 days) the portfolio lost 66.8% and M1 was exceeded on 15 days. Source: Author's own calculations.

4.7 Robustness

Table 13 summarises the 180 specifications of the robustness grid, Figure 4 shows the range of results for each model, and Appendixes B and C list every specification. The main conclusions do not depend on the pre-registered choices. All rejections are counted with the pre-registered one-sided test, and Table 13 also shows the two-sided counts. At the 99% level, the LTCM-style model is rejected in all 18 of its specifications, with exceedance rates between 1.71% and 2.51%, and its exceedances cluster in every one. The RiskMetrics model is also rejected in all 18. The three fat-tailed models are rejected in 15 (historical simulation), 15 (GARCH-t) and 12 (extreme value) of their 18 specifications; with the two-sided test the counts are 15, 13 and 12.

Two choices drive most of the variation: the estimation window and the maturity of the liquidity trade. The durations of the bond trades make almost no difference to any result, because the inverse-volatility weights absorb them, so the 180 specifications amount to about 60 distinct ones. The maturity of the liquidity trade matters more, although part of its apparent effect could come from the sample: the 30-year specifications exclude February 2002 to February 2006, and the 10-year ones do not. With an expanding window, which by the end of the sample contains 35 years of data, the GARCH-t model's exceedance rate is 0.94% with the 10-year trade, and it passes both tests, but 1.32% with the 30-year trade, where it is rejected. A post-hoc check runs the 10-year specifications on the same 6,651 days as the 30-year ones. For the LTCM-style model the gap narrows, from 1.71% against 2.47% to 1.88% against 2.47%, so part of it was a sample effect. For GARCH-t it does not: the rate is 0.92% with the 10-year trade against 1.32% with the 30-year trade, a gap of 0.40 points against 0.38 before, so the maturity effect for this model survives in full. The same check also shows how close to the line the primary result lies: on these days the 10-year GARCH-t specification with the five-year window is not rejected (1.19%, one-sided $p = 0.073$). With a two-year window the GARCH-t rate lies between 1.27% and 1.43%. At the 95% level, the pattern of the primary specification is repeated: the normal models are never exceeded significantly too often, although the RiskMetrics model's rate lies slightly above 5% in all 18 specifications (5.03% to 5.43%); the LTCM-style model is exceeded significantly too rarely in 6 of 18 specifications under the two-sided test; and the fat-tailed models are exceeded slightly too often. A researcher free to choose among these specifications after seeing the results could have reported a GARCH-t model that passes its backtest, either by choosing the expanding window with the 10-year trade or, in the primary specification, simply by choosing the two-sided test. The pre-registered specification and test do not pass it, and they are the ones on which the conclusions rest, but the margin is narrow.

Table 13. Summary of the 180 robustness specifications.

VaR level	Model	N	Primary rate (%)	Median rate (%)	Range (%)	Kupiec rejects, one-sided	Kupiec rejects, two-sided	Clustering rejects
99%	M1 Normal, 5-year	18	2.26	2.33	1.71 – 2.51	18 / 18	18 / 18	18 / 18
	M2 EWMA (RiskMetrics)	18	2.25	2.48	2.23 – 2.80	18 / 18	18 / 18	9 / 18
	M3 Historical simulation	18	1.41	1.45	0.97 – 1.61	15 / 18	15 / 18	16 / 18
	M4 GARCH-t filtered HS	18	1.23	1.27	0.94 – 1.43	15 / 18	13 / 18	3 / 18
	M5 Extreme value (GPD)	18	1.29	1.38	0.98 – 1.52	12 / 18	12 / 18	15 / 18
95%	M1 Normal, 5-year	18	4.43	4.67	3.59 – 4.92	0 / 18	6 / 18	18 / 18
	M2 EWMA (RiskMetrics)	18	5.07	5.25	5.03 – 5.43	0 / 18	0 / 18	0 / 18
	M3 Historical simulation	18	5.55	5.65	4.68 – 6.15	15 / 18	12 / 18	18 / 18
	M4 GARCH-t filtered HS	18	5.64	5.55	5.29 – 6.04	15 / 18	9 / 18	0 / 18
	M5 Extreme value (GPD)	18	5.55	5.65	4.68 – 6.15	= M3	= M3	18 / 18

Note: Each model and level has 18 specifications (36 per model) = 3 estimation windows (2 years, 5 years, expanding) × 3 duration pairs (7/6, 10/8, 13/10) × 2 liquidity-trade maturities (10, 30 years). Nominal rates: 1% and 5%. The one-sided Kupiec test (too many exceedances) is the pre-registered test; the two-sided test also rejects models with too few exceedances. All tests at 5%. Clustering rejects = Christoffersen (1998) independence test. At the 95% level M5 is identical to M3 by construction. The duration pairs barely change any result, so the 180 specifications amount to about 60 distinct ones. In the 30-year specifications, 18 February 2002 to 9 February 2006 is excluded. Source: Author's own calculations.

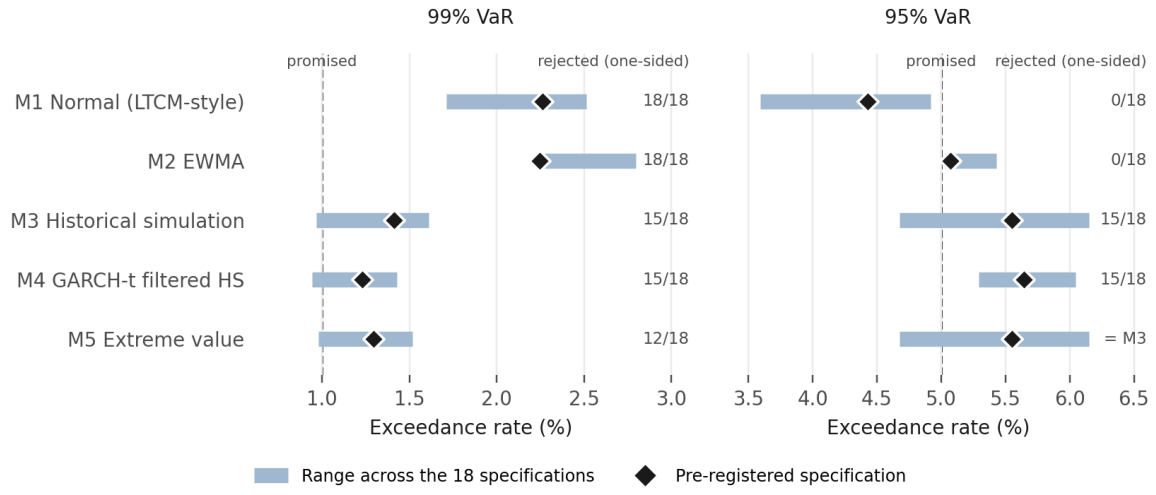


Figure 4. Range of exceedance rates across the 18 robustness specifications of each model, at the 99% and 95% levels. The bar runs from the lowest to the highest rate, the diamond marks the pre-registered specification and the dashed line the promised rate. The numbers on the right count the specifications rejected by the pre-registered one-sided Kupiec test at the 5% level; at 95%, M5 is identical to M3. Each specification is listed in Appendixes B and C. Figure produced using Matplotlib (Hunter, 2007).

4.8 Verification

Two checks were run on the primary results. First, the exceedance count of the LTCM-style model was recomputed with a separate program that rebuilds each day's forecast from the

raw trade returns and weights without using the forecasting code. It finds the same 7,654 days and the same 173 exceedances. Second, a test for look-ahead bias replaced every trade return and portfolio weight after 31 December 1998 with random numbers and recomputed all forecasts for 1998. Had any model used future information, its 1998 forecasts would have changed. In a first version of this check the 1998 forecasts of the GARCH-t model differed by up to 2.8×10^{-7}, although its inputs were identical. The cause was the memory layout of the corrupted weight matrix, which made the linear-algebra library add the same numbers in a different order and so change the last binary digit of the portfolio returns; the GARCH optimiser amplified these differences of about 10^{-17}. With the input arrays given a fixed layout, the 1998 forecasts of every model are identical to the last digit, and so are the exceedance days.

5. Discussion

This chapter interprets the results. It asks whether LTCM's risk model could have seen 1998 coming, what a better model would and would not have achieved, and what the results imply for the management of convergence trades. It closes with the limitations of the study.

5.1 Could the model have seen 1998 coming?

The answer suggested by this study is no, and the reason is not that the crisis came from nowhere. The LTCM-style model fails its 30-year backtest by a wide margin, in calm years as well as in crises, and it describes the 1998 crisis as an event with a probability of less than one in a million. Yet the portfolio's own history already contained a larger one-day loss, in October 1997, and the losses of 1998 were exceeded several times in the following two decades. A model that treats a 12-standard-deviation day in its own sample as a curiosity rather than as information will be surprised again. In this sense the 1998 crisis was not a "ten-sigma event"; it was an ordinary tail event for a fat-tailed, short-volatility portfolio, measured with a model that assumed thin tails.

5.2 What a better model would and would not have achieved

The alternative models improve on the LTCM-style model in exactly the ways the literature predicts. Models that allow for fat tails reduce the number of exceedances, and the GARCH-t model, which also reacts to volatility within days, comes close to the promised rate and removes the clustering. Its forecast roughly doubled at the end of August 1998. A risk manager using it would have been told to cut positions within days of the worst losses.

Two qualifications limit this conclusion. First, every model understates the size of losses beyond the VaR, and the best pre-registered model is still rejected, if narrowly; only the GARCH-filtered extreme-value model added after the results passes the 99% VaR backtest, and it fails at 95%. The improvement is real, but it is one of degree. Second, and more importantly, a model that reacts to volatility after it has risen tells a leveraged arbitrageur to sell at the moment when everyone else is selling. This is the mechanism described by Shleifer and Vishny (1997) and Brunnermeier and Pedersen (2009). For a small investor, a faster model reduces losses. For a fund the size of LTCM, holding positions that other arbitrageurs had copied (MacKenzie, 2003), a faster model might have brought forward the forced sales that deepened the crisis. No daily risk model can fix a position that is too large to exit.

The synthetic portfolio also illustrates the difference between a trade and a fund. Summed over the whole of 1998, the portfolio lost only 1.2%, and in 1999 it gained 46%, because the spreads that widened in August and September narrowed again once the panic passed (with compounding, −3.7% and +56.7%). The trades were, in a sense, right in 1998, but only for an investor who kept the positions through the losses. The simulation keeps its capital constant, and over 2007–2008 the same trades lost 76% from peak to trough even with compounding. LTCM did not survive long enough to collect.

5.3 Lessons for the risk management of convergence trades

Three lessons follow. First, the tails of short-volatility and spread trades must be modelled directly, not inferred from their standard deviation: the 95% results show that a normal model can look conservative for moderate losses while being badly wrong for extreme ones. Second, the diversification between trades that share an exposure to liquidity and credit is unreliable. The correlations of the three trades rose from almost zero to 0.42 in 1998, and in the five years before 2008 and before 2020 they averaged 0.21 to 0.27, well above the near-zero level an investor would have estimated before 1998. Correlations estimated in calm periods should be treated as a lower bound. Third, leverage converts a statistical failure into an economic one. A portfolio that loses 26% of its capital at 15% target volatility survives the 1998 window; the same positions at LTCM's leverage do not. The 2008 drawdown of 76% shows that even the 15%-target portfolio, with long bond positions of six to eight times capital but far less leverage than LTCM, would not have been survivable for a fund with outside investors.

5.4 Limitations

The study has several limitations. The first and most important is that the synthetic portfolio is not LTCM's book. LTCM held around 60,000 positions that were never published, and the pre-registered validation check shows that the portfolio captures its August and September losses but not those of May and June. Second, the short-volatility trade uses the 30-day VIX, whereas LTCM sold options with maturities of about five years, whose implied volatility moves less. Third, the swap-spread trade and the equity pairs that caused a large part of LTCM's losses cannot be tested for 1998 with free data. Fourth, the bond trades use assumed rather than measured durations, and the fitted off-the-run curve contains small fitting errors that add noise to the liquidity trade; the robustness grid shows that the conclusions do not depend on the durations. Fifth, the portfolio is never forced to sell and its trades do not move prices, whereas LTCM's size and the front-running it suffered made its losses larger than any price-taking simulation can show. Sixth, returns are summed on constant capital, which implicitly recapitalises the portfolio after losses; the compounded results are reported in Section 4.1. Seventh, the comparison of correlations is subject to the volatility bias described by Forbes and Rigobon (2002). Finally, the pre-registration plan was not lodged with a public registry, and some choices listed in Appendix A were made after results had been seen.

6. Conclusion

This study asked whether a risk model of the kind LTCM used would have warned of the 1998 losses on LTCM-style trades. Under a pre-registered design, with three synthetic trades built from free public data and five risk models backtested on 7,654 days from 1995 to 2025, the answer is that it would not. The LTCM-style model, a normal distribution estimated on five years of data, was exceeded 173 times where 77 were expected, and it assigned the 1998 crisis a probability of less than one in a million, although its own estimation sample contained a larger one-day loss. The primary pre-registered test rejects it decisively.

Better models help but do not solve the problem. A GARCH model with fat-tailed errors comes close to its promised exceedance rate and reacts within days to a crisis, but it is narrowly rejected by the pre-registered one-sided test, and a GARCH model with an extreme-value tail, added after the results were seen, passes the 99% VaR backtest but not the 95% one. All models understate the size of the worst losses. The trades' correlations rose from almost zero to 0.42 in 1998, and in the five years before 2008 and before 2020 they averaged 0.21 to 0.27, well above their level before 1998. The synthetic portfolio captures LTCM's crash in August and September 1998 but not its earlier losses, a reminder that the study tests trade types and a risk method rather than the fund itself.

The broader lesson is the one LTCM's partners drew too late. A risk model that describes a portfolio by its average behaviour will fail precisely when the portfolio is built to collect small, steady gains in exchange for rare, large losses. For such portfolios, the tail is not a detail of the risk model; it is the risk. Future work could extend the analysis to the swap-spread and equity-pair trades with commercial data, and ask whether the same models would have warned of the later failures of leveraged relative-value funds.

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Appendixes

Appendix A. Deviations from the pre-registration plan.

Change	Reason	Results seen before the change?
The sample ends on 31 May 2025 rather than on the last date in the data files	Author's choice of sample period	No
Bond spreads are carried forward for up to three days when the stock market is open and the bond market closed	Implementation detail: keeps the trades on a common calendar	No
The random seed used for simulations and bootstrap draws was changed from the value in the plan	Author's choice. With the seed in the plan: Z2 p-values M1 < 0.001, M2 < 0.001, M3 < 0.001, M4 0.010, M5 0.004; H1 cumulative probabilities M3 5.9×10^{-5}, M4 2.4×10^{-4}, M5 7.1×10^{-4}; H4 p-values 1998 < 0.001, 2008 0.659, 2020 0.585. No label changes	Yes
30-year specifications: a first run kept February 2002 to February 2006, because the data file contains values for those years, although FRED describes the series as discontinued over that period. The values do not measure an on-the-run 30-year bond, so the years are now excluded, as the plan specified	Correction back to the plan; the 30-year specifications are secondary	Yes
Supplementary model M6 (GARCH-filtered extreme value, McNeil and Frey, 2000) added in the primary specification	The pre-registered M5 is the unconditional version; M6 is the standard conditional benchmark	Yes
Forbes–Rigobon adjusted correlations, GARCH-dynamic cumulative probabilities, and the 10-year specifications rerun on the same days as the 30-year ones, added as checks	Robustness of H4, H1 and the maturity comparison to known biases	Yes
The swap spread is no longer carried forward for three days after the swap series ends (last observation 28 October 2016)	Correction: carry-forward is meant for bond-market holidays only	Yes
The TED spread, listed in the plan to describe funding stress, is not used	Not needed for any test; the series ends in January 2022	No
H3 is reported as not supported but labelled Inconclusive	The plan defined no 'Contradicted' rule for H3	Yes (labelling only)

Note: No other change was made to the plan. Source: Pre-registration plan, deviation log.

Appendix B. Exceedance rates (%) of all 90 specifications at the 99% level.

Window	Durations	Liquidity maturity	Days	M1	M2	M3	M4	M5
2 years	7 / 6	10	7,654	2.19^{***}	2.23^{***}	1.40^{***}	1.28^{***}	1.19^{*}
2 years	7 / 6	30	6,651	2.41^{***}	2.71^{***}	1.47^{***}	1.43^{***}	1.52^{***}
2 years	10 / 8	10	7,654	2.21^{***}	2.25^{***}	1.41^{***}	1.27^{**}	1.19^{*}
2 years	10 / 8	30	6,651	2.41^{***}	2.80^{***}	1.47^{***}	1.43^{***}	1.52^{***}
2 years	13 / 10	10	7,654	2.22^{***}	2.25^{***}	1.42^{***}	1.27^{**}	1.19^{*}
2 years	13 / 10	30	6,651	2.41^{***}	2.80^{***}	1.47^{***}	1.41^{***}	1.52^{***}

Window	Durations	Liquidity maturity	Days	M1	M2	M3	M4	M5
5 years	7 / 6	10	7,654	2.26***	2.23***	1.41***	1.24**	1.29***
5 years	7 / 6	30	6,651	2.41***	2.71***	1.61***	1.26**	1.50***
5 years†	10 / 8	10	7,654	2.26***	2.25***	1.41***	1.23**	1.29***
5 years	10 / 8	30	6,651	2.42***	2.80***	1.61***	1.28**	1.50***
5 years	13 / 10	10	7,654	2.26***	2.25***	1.41***	1.23**	1.28***
5 years	13 / 10	30	6,651	2.42***	2.80***	1.61***	1.28**	1.50***
Expanding	7 / 6	10	7,654	1.71***	2.23***	0.98	0.94	0.98
Expanding	7 / 6	30	6,651	2.44***	2.71***	1.55***	1.32***	1.47***
Expanding	10 / 8	10	7,654	1.71***	2.25***	0.97	0.94	0.98
Expanding	10 / 8	30	6,651	2.47***	2.80***	1.55***	1.32***	1.47***
Expanding	13 / 10	10	7,654	1.71***	2.25***	0.97	0.94	0.98
Expanding	13 / 10	30	6,651	2.51***	2.80***	1.55***	1.32***	1.47***

*Note: Pre-registered one-sided Kupiec test (too many exceedances) against the nominal rate of 1%: *** p < 0.01, ** p < 0.05, * p < 0.10. † Pre-registered primary specification. 30-year specifications exclude 18 February 2002 to 9 February 2006. Source: Author's own calculations.*

Appendix C. Exceedance rates (%) of all 90 specifications at the 95% level.

Window	Durations	Liquidity maturity	Days	M1	M2	M3	M4	M5
2 years	7 / 6	10	7,654	4.64	5.03	5.47**	5.47**	5.47**
2 years	7 / 6	30	6,651	4.84	5.40*	5.94***	6.04***	5.94***
2 years	10 / 8	10	7,654	4.66	5.07	5.46**	5.47**	5.46**
2 years	10 / 8	30	6,651	4.90	5.41*	5.94***	6.04***	5.94***
2 years	13 / 10	10	7,654	4.66	5.10	5.46**	5.47**	5.46**
2 years	13 / 10	30	6,651	4.90	5.43*	5.95***	6.04***	5.95***
5 years	7 / 6	10	7,654	4.43	5.03	5.55**	5.63***	5.55**
5 years	7 / 6	30	6,651	4.84	5.40*	6.15***	5.77***	6.15***
5 years†	10 / 8	10	7,654	4.43	5.07	5.55**	5.64***	5.55**
5 years	10 / 8	30	6,651	4.89	5.41*	6.15***	5.77***	6.15***
5 years	13 / 10	10	7,654	4.44	5.10	5.55**	5.63***	5.55**
5 years	13 / 10	30	6,651	4.92	5.43*	6.15***	5.77***	6.15***
Expanding	7 / 6	10	7,654	3.59	5.03	4.69	5.29	4.69
Expanding	7 / 6	30	6,651	4.68	5.40*	5.74***	5.47**	5.74***
Expanding	10 / 8	10	7,654	3.59	5.07	4.68	5.30	4.68
Expanding	10 / 8	30	6,651	4.72	5.41*	5.76***	5.46**	5.76***
Expanding	13 / 10	10	7,654	3.61	5.10	4.70	5.30	4.70
Expanding	13 / 10	30	6,651	4.75	5.43*	5.76***	5.46**	5.76***

*Note: Pre-registered one-sided Kupiec test (too many exceedances) against the nominal rate of 5%: *** p < 0.01, ** p < 0.05, * p < 0.10. † Pre-registered primary specification. 30-year specifications exclude 18 February 2002 to 9 February 2006. M5 is identical to M3 at this level by construction. Source: Author's own calculations.*

Appendix D. Data files and sources.

File	Source	Content used	Downloaded
F-F_Research_Data_Factors_daily.csv	Kenneth French Data Library	Daily market excess return and risk-free rate	16 June 2025

File	Source	Content used	Downloaded
feds200628.csv	Federal Reserve Board (Gürkaynak, Sack and Wright, 2007)	Fitted par yields SVENPY10 and SVENPY30	16 June 2025
daily.csv	FRED, Federal Reserve Bank of St. Louis	DBAA, DGS10, DGS30, DSWP10 (other columns in the file are not used)	17 June 2025
daily,_close.csv	FRED, Federal Reserve Bank of St. Louis	VIXCLS	17 June 2025

Note: All series are used up to 31 May 2025. Moody's yield series may not be redistributed, so the code downloads them from FRED rather than including them. Revisions by the data providers may cause small differences in later downloads. The full code reproduces every table and figure. Source: Author's own calculations.